Data Science Capstone Project Initial Data Look and Database Setup

Course: MSDS640 Data Science Capstone · Dataset: UCI Adult Census Income (1994 U.S. Census) · Erik Herb

Problem motivation and project plan

What my data is

The dataset is the UCI Adult Census Income data from the 1994 U.S. Census. It has about 32,500 rows and 15 raw columns, and each row is one person. The target is income, either over 50K or 50K and under a year, and the task in the later milestones is to predict which side of that line a person falls on from their demographic, education, and job information. I keep age, workclass, education as a number, marital status, occupation, relationship, race, sex, capital gain, capital loss, hours per week, and native country as predictors. I drop the final sampling weight because it is a census weight and not a fact about the person, and I drop the text education column because it repeats the numeric one.

What database I am using and why

I am using SQLite. It is a lightweight, file based database that lives in a single adult.db file, so there is no server to set up and I can commit the whole database straight into the repo. Pandas reads from it directly with a normal SQL query, which is all I need for a static dataset of this size worked on by one person. Once the database is built it becomes the source of truth, and everything after that reads the data back out of it instead of the raw CSV.

Context and background

The point of this project is not just to get a high accuracy score. It is to understand what the model is doing and who it works for. To make that concrete I frame the use case as a lender or screening tool that flags likely high earners, where the features driving the decision and the fairness across groups matter more than raw accuracy.

Hypotheses

The project is built around three hypotheses. Hypothesis 1 is that education level is a stronger driver of the over 50K prediction than hours worked. Hypothesis 2 is that XGBoost is the fairest of the three models toward true high earners across nativity, meaning it has a smaller US-born versus foreign-born recall gap on real high earners than logistic regression and the MLP. Hypothesis 3 is that XGBoost classifies better than both the logistic regression baseline and the MLP on PR-AUC and F1.

Experiment plan

This milestone is the groundwork: build the database, pull the data, clean it, and do an initial look. In the next milestones I add feature engineering, handle the class imbalance, train the three models, and evaluate them. I lead with PR-AUC and recall rather than accuracy because the target is imbalanced. For Hypothesis 1 I compare feature importance and coefficients on the trained model, for Hypothesis 2 I split recall by nativity and measure the gap for each model, and for Hypothesis 3 I compare the models on held out PR-AUC and F1 with significance tests.

ML methods

The three models are logistic regression as the baseline, XGBoost as the tree model, and a multilayer perceptron as the neural network.

I also use a decision tree, but not as a fourth model competing with those three. I use it as an interpretability tool, which is what my professor suggested. A tree estimates feature importance with every feature competing for the same splits at once, which is exactly what Hypothesis 1 needs and which a one-feature-at-a-time view cannot give me. A shallow tree is also readable on its own, and a shallow tree fit on a complex model's predictions gives a readable approximation of what that model is doing. That is the basis for the explainability note in the final report's model comparison section.

Database setup

I read the raw CSV once and write it into the SQLite database, telling pandas to treat the question mark placeholders as missing values and to trim the leading spaces in the text fields. This saves adult.db with 32,561 rows, and that is the file I commit to the repo. I then pull the data back out with a SQL query and save a CSV copy, so everything after this runs on the dataframe that came out of the database. The pulled data has the same shape as the raw file, and the missing values come back as NaN because they were stored as nulls.

Initial data look

Before cleaning anything I check the shape, the column types, and the numeric summary. This is where the missing values first show up, since three columns come back with fewer non-null rows than the total. The numeric summary shows age averages about 38.6, education is around 10 years, hours per week clusters at 40, and capital gain and capital loss are almost always zero with a few very large values.

Data cleaning

The order of the cleaning steps matters. I first take a snapshot of the missing values, the duplicates, and the target balance, which shows 1,836 missing in workclass, 1,843 in occupation, and 583 in native country, plus 24 exact duplicate rows, and an income split of about 75.9 percent under 50K and 24.1 percent over 50K.

I fill the missing categorical values with an explicit Unknown category instead of dropping the rows, so I keep those people in the data. I then drop the 24 exact duplicate rows while the sampling weight is still in the table, on purpose, because that weight is close to unique per row and dropping it first would make a lot of different people look identical and get removed by mistake. After that I drop the sampling weight and the text education column and rename the columns so the dots become underscores. The cleaned data has 32,537 rows, 13 columns, and no missing values left.

Data exploration

The histograms of the numeric columns show capital gain and capital loss are very right skewed, hours per week clusters hard at 40, age is spread out and roughly bell shaped, and education is discrete and centered around 10 years. A boxplot of education by income and a bar chart of the over 50K rate by education both show the higher earners sit higher on the education scale and the rate climbs steadily as education increases, though there is enough overlap that education alone cannot predict income perfectly.

The class balance chart confirms the target is imbalanced at about 75.9 percent under 50K versus 24.1 percent over 50K, which is why accuracy on its own is misleading and I lead with PR-AUC instead. The correlation heatmap of the numeric features shows only small off-diagonal values, so there is no real multicollinearity and I do not need to drop any numeric feature. Finally I start the fairness setup by grouping native country into US versus non-US, keeping Unknown separate, and looking at the over 50K rate for each group. US-born people earn over 50K at about 24.6 percent and foreign-born people at about 18.7 percent, and that base-rate gap is exactly what Hypothesis 2 will later audit.

How I used an LLM for this assignment

I used LLMs to help me find the dataset for this project as well as develop a good research question and fine tune how I would properly test my question using different models. I also used LLMs to help me troubleshoot different bugs in my code and develop some explanations for the outputs I got so I could correctly interpret my results from the visuals I made.

For this revision I used an LLM to work through my professor's feedback about using a decision tree for feature importance. I wrote and edited the analysis myself and confirmed every number by running the notebook top to bottom.

Finally, I used LLMs to compare what I had completed for the assignment against what was required in the instructions to make sure I was not missing anything needed to be submitted.